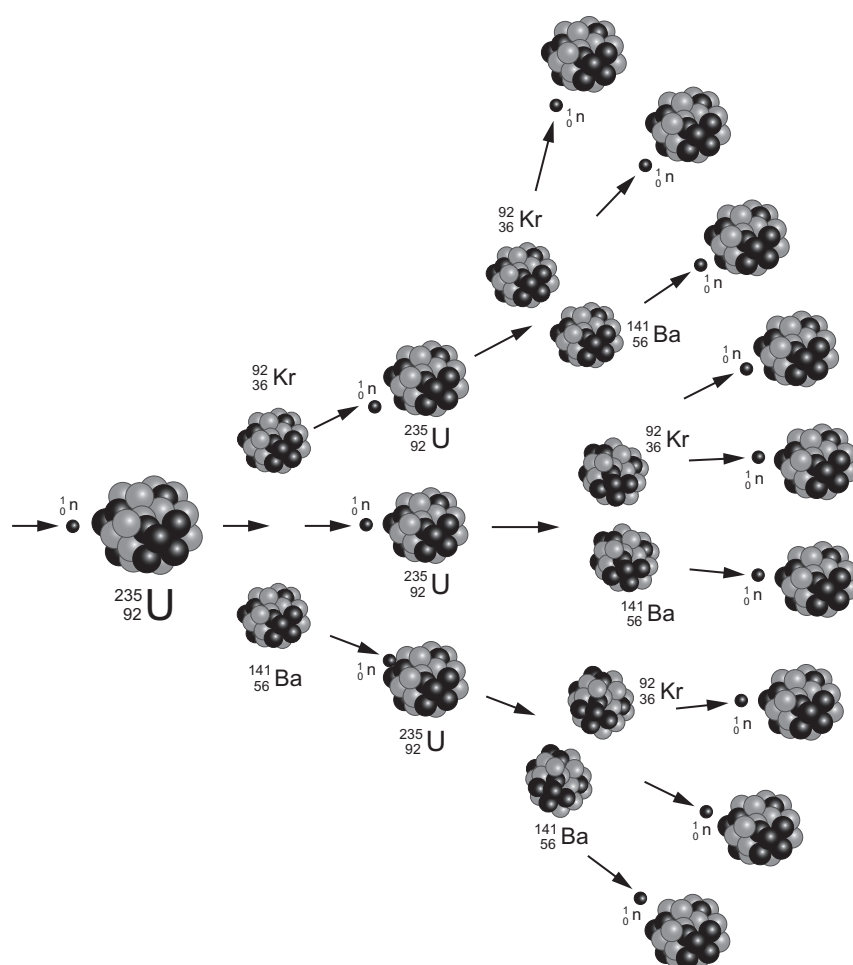


4. Information about two natural isotopes of uranium is given in the table.

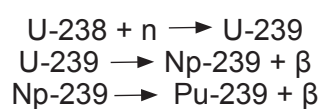
Isotope	Symbol	Type of decay	Half life (years)	Product
uranium-235	${}^{235}_{92}\text{U}$	alpha	7×10^8	thorium-231
uranium-238	${}^{238}_{92}\text{U}$	alpha	4.5×10^9	thorium-234

One type of fission reaction involving uranium-235 is shown in the diagram.



Another type of reaction occurs in a fast breeder fission nuclear reactor. Uranium-238 (U-238) can be used to generate plutonium-239 (Pu-239), which also acts as a reactor fuel supply. An intermediate stage involves the production of neptunium-239 (Np-239).

The reaction follows the three stages below:



- (a) (i) State what is meant by the term half-life. [2]

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- (ii) The isotopes of uranium in the table have been in existence since the Earth was formed. One estimate of the Earth's age is 4.2 billion years. Calculate the fraction of uranium-235 that remains on Earth today. [3]

Fraction =

- (b) (i) Explain how the uncontrolled chain reaction shown in the diagram is prevented in a nuclear reactor. [3]

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- (ii) Describe the structure of a nucleus of $^{141}_{56}\text{Ba}$ which is one of the fission products. [2]

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- (c) Complete the following nuclear equations showing how uranium-238 generates plutonium. [4]

